

Obstacle-Avoiding AUTONOMOUS CAR ROBOT Using Arduino

DR S. RAJAN AND MYTHILI M.

The use of autonomous systems to perform relevant and delicate tasks is fast growing and their application in various fields cannot be over emphasized. But obstacle avoidance is an important task in robotics, as the autonomous robot's aim is to reach the destination without collisions. One type of autonomous robot that can detect obstacles and edges, and takes alternative paths free of obstacles and edges, is a real-time

obstacle avoiding edge detection robot.

This project proposes a robot with intelligence built into it, which

guides it whenever an obstacle comes its way using an algorithm. The design allows the robot to navigate in an unknown environment by

Bill of Material

- | | |
|----------------------------------|----|
| 1. Arduino Uno R3 | -1 |
| 2. Motor driver shield | -1 |
| 3. Wheels | -4 |
| 4. DC gear motors | -4 |
| 5. Servo motor | -1 |
| 6. Ultrasonic sensor | -1 |
| 7. Sensor stand | -1 |
| 8. Lipo battery | -1 |
| 9. Acrylic sheet | -1 |
| 10. Male and female jumper wires | |



Fig. 1: Authors' prototype

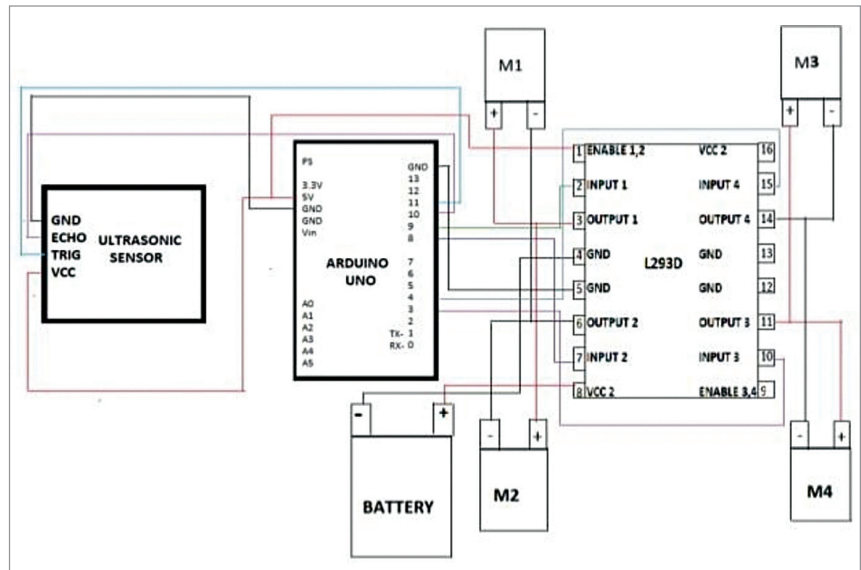


Fig. 2: Block diagram of the obstacle-avoiding robot

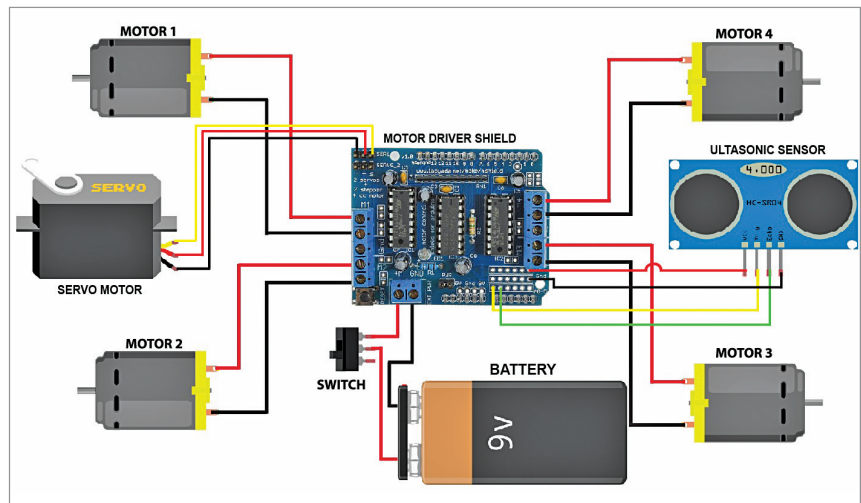


Fig. 3: Circuit diagram of the obstacle-avoiding robot